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# SOUTH AMERICAN PALAEOZOIC PALAEOMAGNETIC RESULTS AND THE WELDING OF PANGAEA

R. THOMPSON
*School of Physics, University of Newcastle upon Tyne, Newcastle upon Tyne, NE1 7RU, England*

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Cambrian, Ordovician and Lower Devonian sediments from northern Argentina have been examined palaeomagnetically. Of the six Cambrian localities investigated, five are shown to have magnetically stable deposits. The Devonian samples, however, are demonstrated to hold two secondary magnetizations and no detectable primary remanence. The five Cambrian and one Ordovician palaeomagnetic pole positions

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| South Tilcara | 52N | 27E | ($A_{95}$ = 23) | Purmamarca vill | 61N | 67E | ($A_{95}$ = 26) |
| North Tilcara | 49N | 24E | ($A_{95}$ = 23) | Purmamarca | 5N | 39E | ($A_{95}$ = 14) |
| Abra de Cajas | 2N | 28E | ($A_{95}$ = 50) | Salta (Ordovician) | 31N | 13E | ($A_{95}$ = 40) |

are used in conjunction with three previously determined Lower Palaeozoic palaeomagnetic pole positions to suggest a new South American Lower Palaeozoic polar wander path, which with palaeomagnetic results from other southern continents forms a common Gondwanan Lower Palaeozoic polar wander path. The relationship of the Gondwanan curve with the North American, European and Asian Palaeozoic polar wander paths is discussed in terms of the welding of Pangaea.

## 1. Introduction

Palaeozoic palaeomagnetic collections from South America were made in 1956 and 1958. The results of these surveys suggested that red sediments from the North-west Argentinian provinces of Salta and Jujuy were suitable for more detailed palaeomagnetic investigations [1]. During a field expedition in 1969, 130 hand samples of Cambrian, Ordovician and Lower Devonian sediments were collected. The samples were orientated by use of both magnetic and sun compasses [2]. The palaeomagnetic results from the new collection are discussed. They show that the south palaeomagnetic pole lay further to the north-east of South America in the Cambrian and Ordovician than previously suggested. The South American results fall into a consistent pattern with results from other Gondwanan continents and form a single polar wander curve. This path is contrasted with the paths from Europe, North America and Asia and is compared with the geological records of the Palaeozoic mountain belts.

## 2. South American palaeozoic palaeomagnetic results

### *2.1. Geology*

The Lower Palaeozoic rocks are commonly exposed on large, structurally simple, north-south trending blocks or folds (figs. 1 and 2). The Cambrian sediments lie unconformably on weakly metamorphosed Precambrian, with a basal conglomerate commonly developed. The sediments vary considerably but are frequently quartzitic, medium grained sandstones of a pinkish or light colour. They are assigned to the Cambrian Period simply on their stratigraphical level below Lower Tremadocian faunas, and on the occurrence of inarticulate brachiopods and Skolithos tubes [3]. The Ordovician lies unconformably on the Cambrian and is fossiliferous with typical Ordovician associations of trilobites and brachiopods, and graptolites in alternations of sandstones and shales. The Devonian sequences of Argentina have been recently reviewed by Padula et al. [4]. Lower Devonian samples were

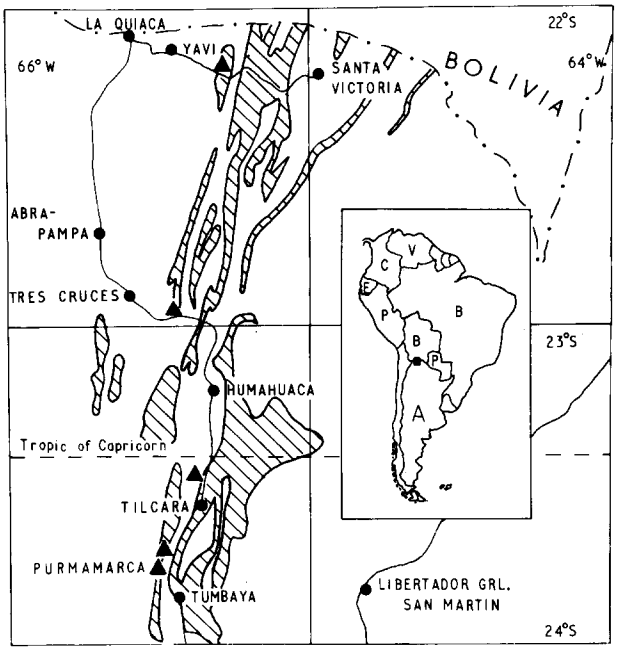



Fig. 1. Collection localities of Cambrian samples. Outcrops of Cambrian strata in N.W. Argentina (shaded).

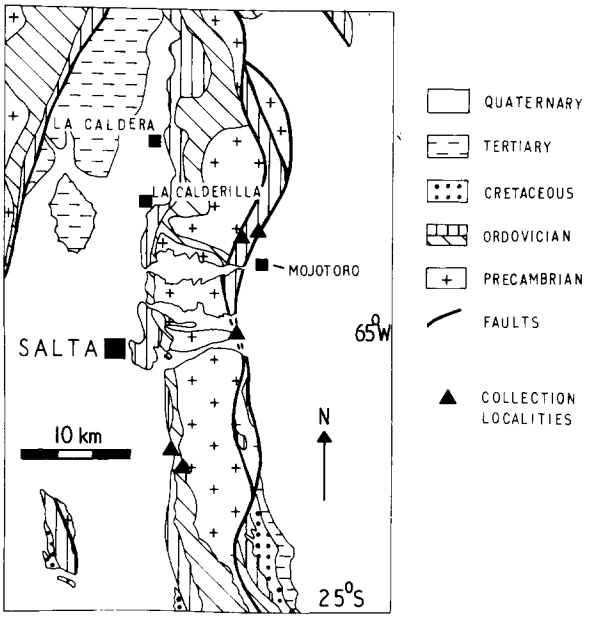



Fig. 2. Collection localities of Ordovician sediments near Salta in N.W. Argentina.

collected from the upper part of the Baritu formation in the Sierra de Zapala, and were of silicified sandstone with a red to reddish-purple colour. The colouration was restricted to certain horizons and occurred in both the cement and small cracks and joints within the beds. Other layers were of white, massive, sandstones with cross-bedding which showed that the beds were the right way up. The strata occurred in shallow anticlines and synclines with inclinations of up to 10° on the limbs of the folds.

### 2.2. *Laboratory techniques*

All the hand samples were returned to Newcastle upon Tyne where, whenever possible, three 2.5 cm diameter cores were drilled from each sample. One 2.5 cm high cylinder was then cut from each of the cores. Measurements of the magnetic remanence were made on a PAR spinner magnetometer or the No. 1 sensitive astatic magnetometer [2]. Step thermal cleaning of pilot specimens was carried out to determine if weak secondary magnetizations could be removed by thermal partial demagnetization [2]. After each step of cleaning the initial susceptibility of the specimens was measured to determine whether chemical changes had occurred during the thermal treatment. Fisherian statistics [5] have been used for the calculation of mean directions and their errors, and unless stated otherwise all results have been corrected for the orientation of the bedding.

### 2.3. *Cambrian*

Cambrian sediments were collected from six localities (fig. 1). The cleaned magnetic remanence was considered to be stable at all the localities except Tres Cruces. The five stable mean results are tabulated in table 1 and the palaeomagnetic poles plotted in fig. 7.

#### 2.3.1. *Tilcara (south)*

Light-red, massive Cambrian sandstones lay unconformably on Precambrian rocks about 16 km south of Tilcara. The sediments were particularly well exposed and eleven hand samples were collected from a section about 30 metres thick where the beds were dipping uniformly at 50°. The NRM had a geometrical mean intensity of $1.9 \times 10^{-6}$ G (Gauss). Both normal and reverse directions of magnetization were

Table 1
Cambrian and Ordovician palaeomagnetic results

| Locality | Orientation | Temp | $N(n)$ | Dec | Inc | $\alpha_{95}$ | Lat° | Long° | $A_{95}$ |
|---|---|---|---|---|---|---|---|---|---|
| South Tilcara | Present | NRM | 11 (29) | 62 | − 02 | 24 | | | |
| (24°S 66°W) | Palaeo | NRM | 11 (29) | 50 | − 39 | 24 | + 49 | 30 | 32 |
| | Present | 400°C | 11 (29) | 53 | − 02 | 23 | | | |
| | Palaeo | 400°C | 11 (29) | 41 | − 34 | 23 | + 52 | 27 | 23 |
| North Tilcara | Present | NRM | 11 (32) | 00 | − 42 | 18 | | | |
| (24°S 66°W) | Palaeo | NRM | 11 (32) | 25 | − 27 | 18 | + 68 | 12 | 16 |
| | Present | 450°C | 11 (32) | 17 | − 47 | 27 | | | |
| | Palaeo | 450°C | 11 (32) | 44 | − 25 | 27 | + 49 | 24 | 23 |
| Abra de Cajas | Present | NRM | 6 (17) | 92 | + 05 | 42 | | | |
| (22°S 65°W) | Palaeo | NRM | 6 (17) | 92 | − 31 | 44 | + 03 | 45 | 50 |
| | Present | 350°C | 6 (17) | 84 | + 31 | 48 | | | |
| | Palaeo | 350°C | 6 (17) | 87 | − 04 | 50 | + 02 | 28 | 50 |
| Tres Cruces | Present | NRM | 8 (18) | 36 | − 13 | 38 | | | |
| (23°S 66°W) | Palaeo | NRM | 8 (18) | 03 | − 57 | 38 | + 72 | 118 | 40 |
| | Present | 300°C | 8 (18) | 299 | − 55 | 57 | | | |
| | Palaeo | 300°C | 8 (18) | 272 | − 13 | 53 | + 10 | 205 | 51 |
| Purmamarca village | Present | NRM | 9 (24) | 350 | − 08 | 28 | | | |
| (24°S 66°W) | Palaeo | NRM | 9 (24) | 04 | − 09 | 24 | + 71 | 306 | 21 |
| | Present | 350°C | 9 (24) | 353 | − 07 | 33 | | | |
| | Palaeo | 350°C | 9 (24) | 359 | + 08 | 32 | + 61 | 293 | 26 |
| Purmamarca | Present | NRM | 12 (34) | 18 | − 43 | 12 | | | |
| (24°S 66°W) | Palaeo | NRM | 12 (34) | 46 | − 33 | 11 | + 49 | 27 | 11 |
| | Present | 400°C | 12 (34) | 84 | − 59 | 17 | | | |
| | Palaeo | 400°C | 12 (34) | 92 | − 25 | 15 | + 05 | 39 | 14 |
| Salta (Ordovician) | Present | NRM | 20 (57) | 335 | − 05 | 54 | | | |
| (25°S 65°W) | Palaeo | NRM | 20 (57) | 45 | + 10 | 53 | + 36 | 358 | 46 |
| | Present | 350°C | 20 (57) | 337 | − 09 | 51 | | | |
| | Palaeo | 350°C | 20 (57) | 53 | 00 | 45 | + 31 | 13 | 40 |

$N$ Number of samples
$n$ Number of specimens
Dec Declination in degrees east
Inc Inclination (negative, up; positive, down)
$\alpha_{95}$ semi-angle of cone of 95% confidence about mean direction.
Lat Latitude of palaeomagnetic pole (positive north, negative south)
Long Longitude of palaeomagnetic pole in degrees east.
$A_{95}$ semi-angle of cone of 95% confidence about mean pole.

found in the sandstones. (These are the first normal horizons of magnetization of Lower Palaeozoic age to have been found in South America). The normal directions had a very low dispersion. Three pilot specimens were selected, two from the normal group and one with reversed magnetization. The two specimens of normal polarity exhibited properties suggesting high magnetic stability. The reversely magnetized pilot specimen showed a change in direction, particularly between 350°C and 550°C towards an azimuth more anti-parallel to that of the group of normal directions (fig..3c). Thermal cleaning of all the specimens at 400°C slightly reduced the scatter of directions, but two samples still maintained intermediate directions. Mean directions were calculated for the normal and reverse groups separately and combined. The mean combined results gave a south palaeomagnetic position at 52°N 27°E ($A_{95}$ = 23°) (table 1). The south palaeo-

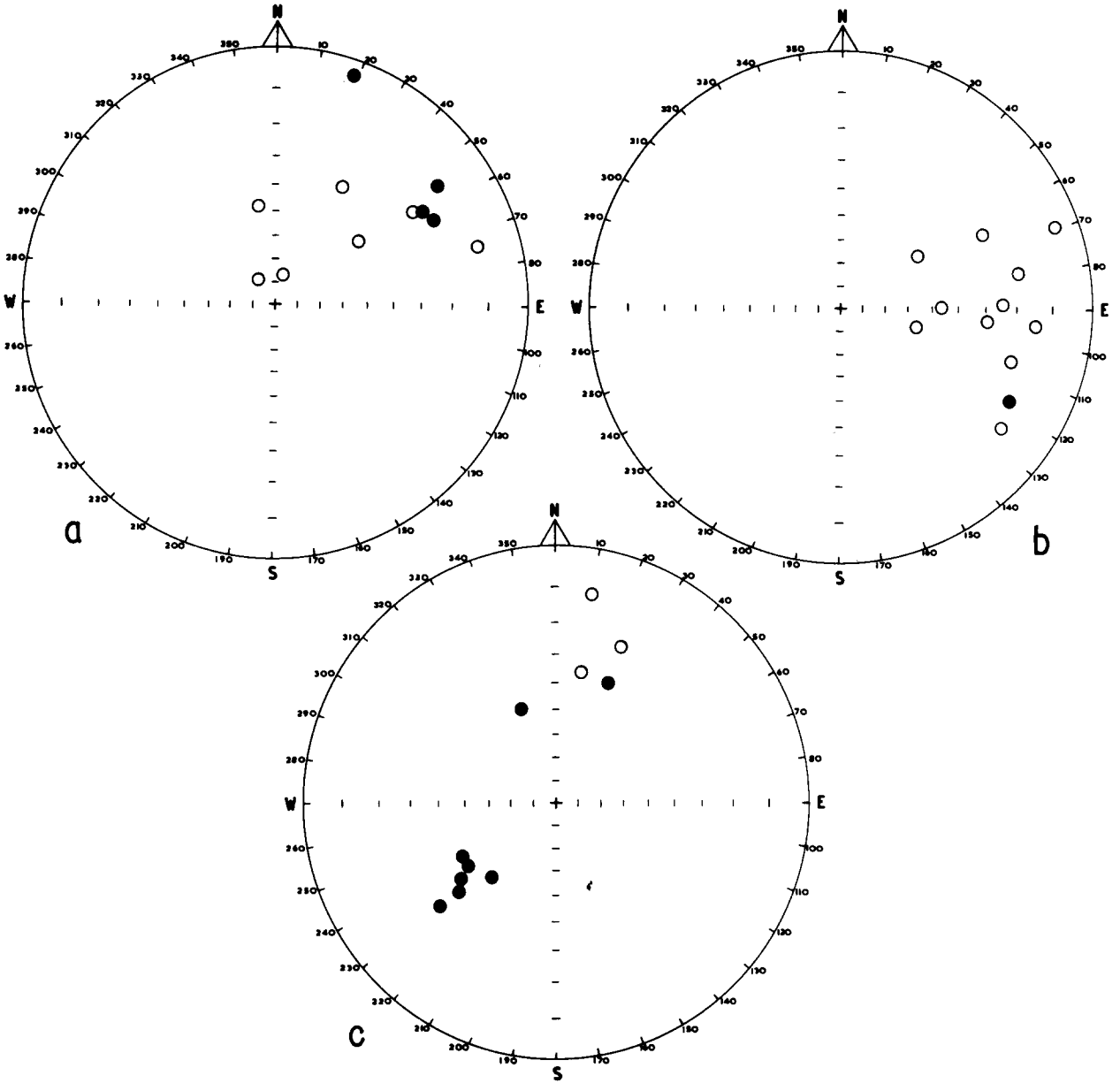



Fig. 3. Stereographic projections of mean sample directions of Cambrian localities: (a) Tilcara (north); (b) Purmamarca; (c) Tilcara (south); after thermal cleaning and correction for the dip of the strata. Open circles, negative (up); closed circles, positive (down).

magnetic pole for the tight group of normal directions lay at 44°N 35°E, $A_{95} = 6°$. (N.B. Because the polar wander path for South America crossed the equator in the Devonian Period, Lower Palaeozoic south palaeomagnetic poles lie in the northern hemisphere.)

*2.3.2. Tilcara (north)*

Ten kilometres to the north of Tilcara where the main road, "Ruta International 9" and the railway line run together, 15 samples were collected from a railway cutting, in pinkish sandstones. The highest beds were

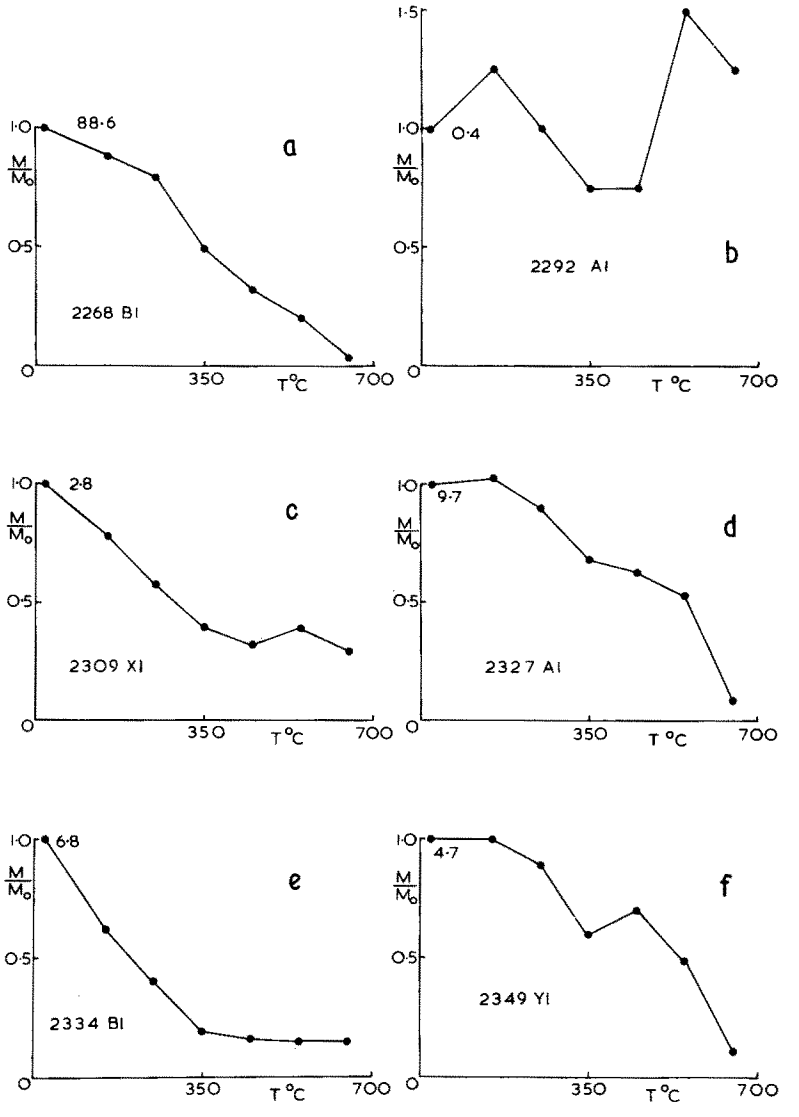



Fig. 4. Cambrian pilot specimens. Thermal decay of remanence with step cleaning. (a) Abra de Cajas; (b) Tres Cruces; (c) Tilcara (north); (d) Tilcara (south); (e) Purmamarca; (f) Purmamarca village. Note contrast between the erratic changes of intensity (indicating an unstable magnetization) in (b) and the gentle decay of the highly stable magnetization of (a) and (d).

whiter and contained fewer pebbly horizons than the lower beds. Five pilot specimens were partially thermally demagnetized in eight steps up to 650°C, and the normalized decrease in intensity for one of the pilots is shown in fig. 4c. After a moderately steep initial fall the curve levelled off between 350°C and 650°C. This behaviour suggested that slightly under half the remanence resulted from grains with a high coercive force, and a temperature of 450°C was selected for bulk partial demagnetization to isolate this high blocking temperature remanence. The directions from the white uppermost horizons remained more scattered than those from the lower beds; so only the eleven samples from the stratigraphically lowest horizons were used in estimating the pole position which lay at 49°N 24°E. The cleaned directions are plotted in fig. 3a and reported in table 1.

*2.3.3. Abra de Cajas*

Cambrian sediments were exposed on the road from Yavi to Santa Victoria, about 25 km east of Yavi (fig. 1) at a height of 4000 m. The beds dipped to the east and cross bedding showed that they were the right way up. Only seven samples were collected, and 17 cylinders were later drilled from them in the laboratory. The collection was made in quartzites of a light pink colour, with a relatively strong mean NRM intensity of $20.5 \times 10^{-6}$ G. The sample mean directions were scattered, $\alpha_{95} = 44^\circ$ (table 1) after the relevant corrections for the dip of the beds had been applied. An example of the decrease of intensity of magnetization for one of the pilot specimens is shown in fig. 4a. Partial thermal demagnetization, at 450°C, of the seventeen specimens did not improve the scatter of directions. However, the magnetization is clearly relatively stable, as shown by the pilot specimens. The mean direction after cleaning is reasonably consistent with the results from the other Cambrian collections.

*2.3.4. Tres Cruces*

38 specimens were drilled from 14 hand samples obtained from Cambrian horizons which were exposed in the valley of the Rio Grande. The intensities of the NRM were low, only two samples having an intensity greater than $0.7 \times 10^{-6}$ G. The pilot specimens did not respond well to demagnetization; note how in fig. 4b the intensity jumped erratically after increasing steps of partial demagnetization temperature, compared with the smoother decreases in intensity of the other pilot specimens shown in fig. 4. Eight quartzite samples were thermally cleaned at 300°C, and the scatter increased from 38° to 53°. This grouping after cleaning was the poorest of any of the Cambrian rocks studied. Attempts to clean the remaining six greenish-grey siltstones led immediately to chemical change and were discontinued. Because of the thermally cleaned results and the behaviour of the pilot specimens, it was concluded that the Cambrian beds near Tres Cruces did not hold a reliable primary remanence.

*2.3.5. Purmamarca village*

Thirteen hand samples were collected from near the school ground at this locality. The Cambrian sediments in this area of the lower part of the Quebrada de Purmamarca have been described by Turner [3]. Nine of the samples were reddish quartzites, while the remaining four were from more shaley, badly fractured and shattered rocks. The latter four samples were difficult to prepare in the laboratory due to their fragile nature. The NRM directions of these four samples were scattered and the remanence became even more diverse with cleaning at 350°C. The nine quartzites were therefore considered alone. Three test specimens showed high stability with cleaning, and results from one are illustrated in fig. 4f. The directions of remanence changed little even at cleaning temperatures of 550°C. The pole position 61°N 293°E, $A_{95} = 26^\circ$ lies considerably to the north and west of the other Cambrian poles reported in this paper (table 1).

*2.3.6. Purmamarca*

The collection was made in fine grained purple siltstones which were well jointed and cut by thin quartz veins at a locality near the junction of the Tilcara to Tumbaya road with the one leading to Purmamarca village. The 12 hand samples supplied 34 cylindrical specimens which were measured on a PAR spinner magnetometer. The NRM magnetization had a mean intensity of $5.4 \times 10^{-6}$ G and gave a lower scatter of directions for the ancient orientation of the rocks compared with the present orientation. The three pilot specimens all showed a relatively steep drop of intensity at the early stages of partial demagnetization below 350°C. Chemical changes were revealed at the last stage of cleaning of 650°C by sharp increases in susceptibility and by changes in direction of the remanent magnetization. In an attempt to separate out the high blocking temperature component, bulk demagnetization was performed at 400°C. The scatter of directions of the sample means increased but was lowest when the directions were corrected for the dip of the beds (fig. 3b).

*2.4. Ordovician*

24 samples were collected near Salta at five separate sites (fig. 2) which lay on a series of tilt blocks. 68 cylinders were cut from the samples and their magnetic remanence was measured on a PAR spinner magnetometer. The NRM showed a high degree of scatter (table 1) and had a low geometrical mean intensity of $1.3 \times 10^{-6}$ G. When referred to the palaeohorizontal the results lay mainly in the north-east quadrant, not unlike previous results from Cambro-Ordovician

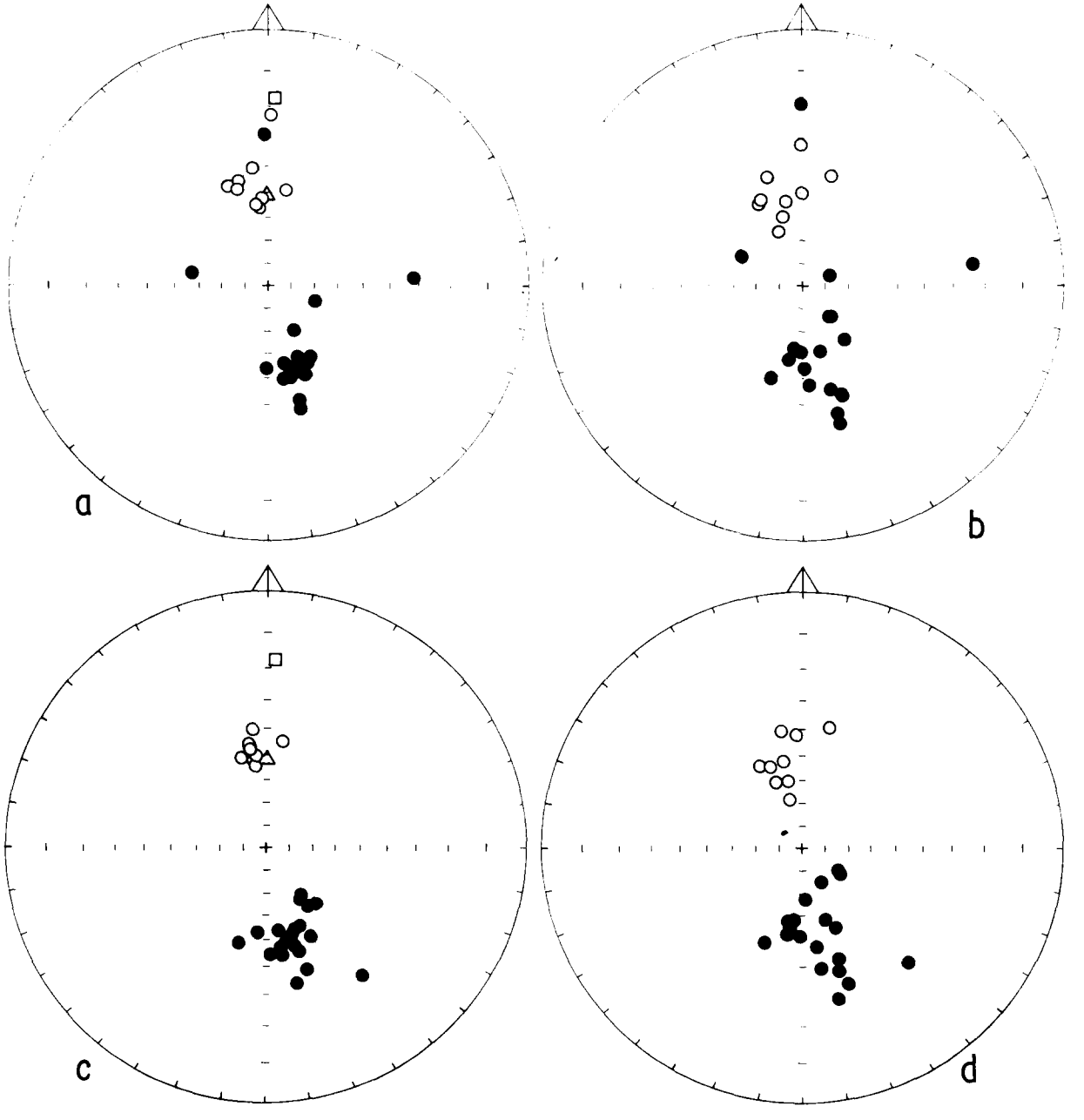



Fig. 5. Devonian (San Pedro) mean sample directions: (a) and (b) NRM; (c) and (d) thermally cleaned at 350°C; (a) and (c) referred to the present horizontal; (b) and (d) to the ancient horizontal. Present magnetic field direction (square), present axial dipole field direction (triangle). Note lower scatter of directions in (a) and (c) (indicating that the dominant magnetization was formed after folding of the beds).

rocks from South America. Eight pilot specimens were subjected to step thermal demagnetization up to 650°C. The directions of remanence did not change greatly at the lower temperatures, and a temperature of 350°C was chosen for bulk cleaning. All four samples from the site a few kilometres south of Mojotoro

Table 2
Lower Devonian palaeomagnetic results

| Orientation | Temp | Polarity | $N(n)$ | Dec | Inc | $\alpha_{95}$ | Lat° | Long° | $A_{95}$ |
|---|---|---|---|---|---|---|---|---|---|
| Present | NRM | Rev | 20 (52) | 156 | + 60 | 12 | − 65 | 341 | 16 |
| Palaeo | NRM | Rev | 20 (52) | 161 | + 61 | 14 | − 67 | 329 | 18 |
| Present | NRM | Norm | 9 (21) | 353 | − 46 | 8 | + 83 | 171 | 8 |
| Palaeo | NRM | Norm | 9 (21) | 350 | − 50 | 9 | + 78 | 163 | 10 |
| Present | NRM | Comb | 29 (73) | 163 | + 55 | 9 | − 71 | 343 | 11 |
| Palaeo | NRM | Comb | 29 (73) | 165 | + 57 | 10 | − 71 | 332 | 12 |
| Present | 350°C | Rev | 20 (52) | 163 | + 50 | 6 | − 73 | 355 | 6 |
| Palaeo | 350°C | Rev | 20 (52) | 167 | + 52 | 8 | − 75 | 339 | 9 |
| Present | 350°C | Norm | 9 (21) | 354 | − 48 | 4 | + 82 | 163 | 4 |
| Palaeo | 350°C | Norm | 9 (21) | 350 | − 52 | 8 | + 77 | 157 | 9 |
| Present | 350°C | Comb | 29 (73) | 166 | + 49 | 4 | − 76 | 353 | 5 |
| Palaeo | 350°C | Comb | 29 (73) | 168 | + 52 | 6 | − 75 | 339 | 6 |

Symbols as for table 1

exhibited high within sample scatter ($\alpha_{95} > 70°$) and were weakly magnetized. In addition it had been very difficult to measure the orientation of the bedding at this site, and so the samples were removed from the calculation of the mean results. The thermally cleaned directions were more tightly grouped than the NRM directions, and when they were referred to the ancient horizontal the scatter decreased from an $\alpha_{95}$ of 51° to 45°. This implied that a small secondary component had been removed during the cleaning at 350°C and that the more stable component was acquired before the tilting of the fault blocks. The mean pole position (table 1) lies to the north and east of the only other Ordovician result from South America (table 3), but not as far to the east as the Cambrian results reported above.

### 2.5. *Devonian*

The 28 samples of Lower Devonian sandstone from near La Mendieta (24°S 65°W) were cut into 76 cylinders and measured on the No. 1 astatic magnetometer at Newcastle upon Tyne. Both normal and reverse directions were recorded and one sample had a particularly anomalous direction. Two samples exhibited both normal and reverse directions. The NRM directions grouped more closely when no correction had been applied for the attitude of the bedding (fig. 5a and b and table 2). Because the directions of remanence of the normal polarity group alone, the reverse polarity group alone and the two groups combined together, all exhibited least scatter when referred to the present horizontal, it was thought that the magnetization was acquired after the folding of the beds. A detailed series of 25 pilot runs was performed, with up to twelve separate stages of step demagnetization. At all stages of thermal cleaning the scatter of directions was lower when the magnetization was referred to the present horizontal rather than the palaeohorizontal. So again the results pointed to a secondary magnetization. The remanence was extremely stable up to temperatures of 675°C (close to the haematite Curie point of 685°C). Heating to 700°C, above the haematite Curie point destroyed the remanence. Bulk thermal cleaning was performed at 350°C and again the directions of remanence grouped more closely when no bedding corrections had been applied to the results. Thus it is most probable that the stable magnetization of these rocks was acquired after the beds had been folded. Comparison of the normal and reverse directions yielded yet another secondary magnetization acquired after folding. This magnetization is revealed by both the difference in shapes of the thermal decay of remanence curves for normally and reversely magnetized specimens (fig. 6) and by the changes in direction with cleaning (table 2) and (fig. 5a and c). For the group of normal magnetization the intensity of remanence decreased with each step of cleaning but for the reverse-

Table 3
Cambrian to Silurian palaeomagnetic pole positions

| Pole | Age | Locality | Lat° | Long° | Ref. |
|---|---|---|---|---|---|
| *South America* | | | | | |
| 1 | C | South Tilcara | + 52 | 27 | a |
| 2 | C | North Tilcara | + 49 | 24 | a |
| 3 | C | Abra de Cajas | + 02 | 28 | a |
| 4 | C | Purmamarca village | + 61 | 293 | a |
| 5 | C | Purmamarca | + 05 | 39 | a |
| 6 | C–O | Red Seds | + 15 | 338 | XII 145 |
| 7 | C–O | Red Seds | + 08 | 320 | XII 144 |
| 8 | O | Salta | + 31 | 13 | a |
| 9 | O | Seds Bolivia | + 04 | 302 | XII 140 |
| *Africa* | | | | | |
| 1 | C | Ntonya | + 28 | 345 | IX 137 |
| 2 | C | Morocco Lavas | + 53 | 34 | VIII 144 |
| 3 | C | Hook Intrusives | + 14 | 336 | IX 132 |
| 4 | O | Table Mountain | + 50 | 349 | X 130 |
| *Australia* | | | | | |
| 1 | C | Antrim Volcanics | – 09 | 340 | XII 148 |
| 2 | C | Hudson | + 18 | 19 | b |
| 3 | O | Jindukin | – 16 | 29 | XII 143 |
| 4 | S | Mugga Mugga | – 80 | 340 | VIII 127 |
| *India* | | | | | |
| 1 | C | Salt Range | – 28 | 32 | XI 85 |
| *North America* | | | | | |
| 1 | C | Bradore | – 09 | 329 | VIII 147 |
| 2 | C | Radcliff | – 10 | 304 | VIII 148 |
| 3 | C | Wichita | – 02 | 327 | X 148 |
| 4 | C | Colorado | – 37 | 302 | c |
| 5 | C | Colorado | – 44 | 280 | c |
| 6 | S | Bloomsbury | – 32 | 282 | IX 125 |
| 7 | S | Rosehill | – 20 | 316 | 4.12 |
| *Antarctica* | | | | | |
| 1 | O | Sor | – 28 | 010 | X 140 |
| *Europe* | | | | | |
| 1 | C | Caerfai | – 26 | 349 | d |
| 2 | C | Hartshill | – 18 | 345 | 2.03 |
| 3 | O | Builth | – 15 | 342 | IX 130 |
| 4 | O | Borrowdale | – 14 | 345 | IX 129 |
| 5 | O | Arenig Scot. | – 11 | 348 | IX 131 |
| 6 | O | Killary | – 09 | 34 | XI 80 |
| 7 | S | Skomer | – 39 | 326 | d |
| 8 | S | Fyne | – 05 | 326 | XII 138 |
| 9 | S | Arrochar | – 08 | 324 | XII 139 |
| 10 | S | ORS Lavas | + 01 | 325 | XII 137 |
| 11 | S | ORS Lavas | – 01 | 301 | IX 124 |
| 12 | S | Ringerike | – 21 | 339 | X 129 |
| *Asia* | | | | | |
| 1 | C | Ur. Lena | + 33 | 310 | X 141 |
| 2 | C | Ur. Lena | + 36 | 299 | X 142 |
| 3 | C | Ur. Lena | + 41 | 307 | X 143 |
| 4 | C | Ur. Lena | + 41 | 306 | X 144 |

Table 3 (continued)

| Pole | Age | Locality | Lat° | Long° | Ref. |
|---|---|---|---|---|---|
| *Asia* | | | | | |
| 5 | C | Verkholnsk | + 34 | 312 | X 145 |
| 6 | O | Makavoski | + 25 | 308 | X 132 |
| 7 | O | Bratsk | + 20 | 300 | X 131 |
| 8 | O | Dolbor | + 21 | 307 | X 133 |
| 9 | O | Mangazeika | + 24 | 311 | X 134 |
| 10 | O | Mangazeika | + 22 | 314 | X 135 |
| 11 | O | Krivaya | + 23 | 308 | X 136 |
| 12 | O | Krivaya | + 27 | 314 | X 137 |
| 13 | O | Redbeds | + 41 | 308 | X 138 |
| 14 | O | Redbeds | + 42 | 311 | X 139 |

Pole numbers referred to in fig. 7.
Ref. a, this paper.
Ref. b, Luck [22].
Ref. c, Lawson and Mutschler [23].
Lat and Long refer to palaeomagnetic pole positions.
Ref. d, Briden et al. [24].
Refs. 2.03, 4.12, Irving [25].
Refs. VIII to XII McElhinny [26–30].
C Cambrian.
O Ordovician.
S Silurian.

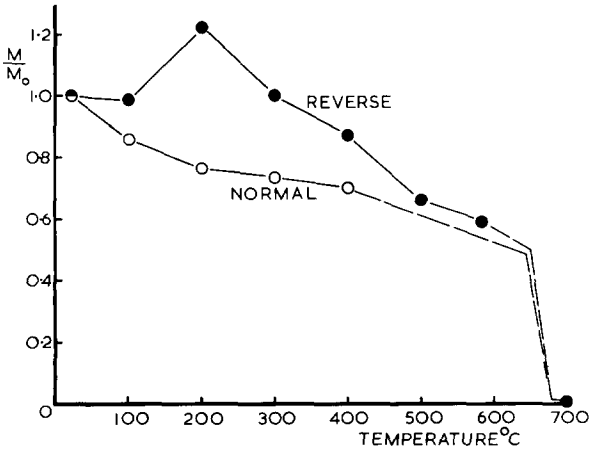



Fig. 6. Mean thermal decay of remanence of Devonian pilot specimens with step cleaning. Note increase of total remanent intensity of the reversely (positively) magnetized specimens at temperatures up to 200°C in contrast to the decrease in intensity of normally magnetized pilot specimens.

ly magnetized group an increase in intensity occurred up to 200°C (fig. 6). Also an increase in inclination with demagnetization for the normal group and a decrease in inclination for the reverse group (fig. 5 and table 2) were produced by the cleaning. All these results combined with the overall decrease in scatter which resulted from bulk demagnetization at 350°C showed that a weaker, additional magnetization, probably in the earth's present field direction (open square in fig. 5a and c) had been removed in the thermal cleaning. The rocks thus contained two secondary components of magnetization (one exceedingly strong) and no detectable primary component, and cannot be used in constructing a South American polar wander path.

Creer [1], section 3.2, reported results from the same locality (samples 513 to 518) which showed little similarity to the author's results. In particular Creer did not report both normal and reverse directions or any stable secondary magnetization. It is considered that the more recent work must be taken to supercede the previous work, which cannot now be used to define part of the Lower Palaeozoic polar wander path for South America. Creer's 'Ordovician' palaeomagnetic pole which was partly based on samples from the above locality is thus not included in table 3.

### 2.6. *Discussion of results*

The six reliable Lower Palaeozoic south palaeomagnetic pole positions obtained from the results just described are listed in table 3 and plotted in fig. 7. The directions have a higher scatter than typical Mesozoic results, but compare favourably with other Palaeozoic results. The distributions of directions about their means are tighter than for the three previous Lower Palaeozoic results from South America, and it is considered that the directions presented above are a true reflection of the Lower Palaeozoic geomagnetic field. Five of the six new palaeomagnetic south pole positions lie to the north and east of the previous Lower Palaeozoic south palaeomagnetic poles from South America (fig. 7). In the following section it is shown that a polar wander curve for the whole of Gondwanaland can be constructed through the region of the new pole positions; and further that this curve contrasts strongly with the Lower Palaeozoic polar wander paths

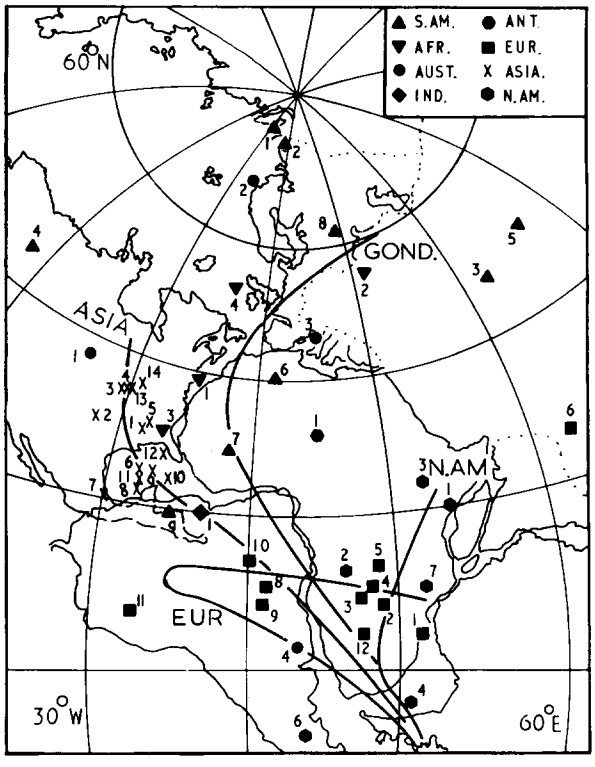



Fig. 7. Worldwide Cambrian to Silurian palaeomagnetic pole positions for a Bullard et al. [7] and Smith and Hallam [8] reconstruction. Poles listed in table 3. Cambrian to Carboniferous polar wander paths for Gondwanaland (Gond), North America (N. Am), Europe (Eur) and Asia are shown.

of North America, Europe and Asia, as would be expected from the geological record of the Hercynian and Appalachian mountain chains.

## 3. The welding of Pangaea

It is now an accepted concept that old mountain chains, within continental regions, especially those containing ophiolite mélanges, reflect belts where ancient continental blocks collided after a period of sea floor spreading. Palaeozoic orogenic belts lie between Laurasia and Gondwanaland (Hercynian and Appalachian mountain chains) and separate Europe from both North America (Caledonian mountain belt) and Asia (Ural mountains). So it is to be expected that each of these four blocks had its own separate polar wander path in the Lower Palaeozoic and further that the ages

of the mountain chains should reflect the times at which the individual polar wander curves coalesce when plotted on a Pangaea reconstruction. Thus considering only the geological record we would expect the European and North American curves to join in the Silurian and then follow a common path to meet the Asian and Gondwanan paths in the Carboniferous. The extent to which palaeomagnetic results follow this pattern is now reviewed.

The quantity and quality of Lower Palaeozoic palaeomagnetic results for the whole world are surprisingly low. A possible explanation of this is the suggestion (e.g. [6]) that the intensity of the geomagnetic field was less in the Lower Palaeozoic than in the late Pre-cambrian, or Upper Palaeozoic and Mesozoic. Cleaned Cambrian to Silurian palaeomagnetic pole positions for the whole world from stable continental areas are listed in table 3 and plotted in fig. 7 on a Bullard et al. [7] and Smith and Hallam [8] reconstruction of Pangaea.

Because there are no orogenic belts of Phanerozoic age which lie within Gondwanaland, only around its edges, it is reasonable to draw a common polar wander path for the whole of Gondwanaland for the Lower Palaeozoic and to compare the South American results directly with those from other southern continents. Prior attempts to define a Lower Palaeozoic polar wander path for Gondwanaland have generally shown such a curve swinging north-east to south-west with increasing age, across the north-west of Africa, e.g. [9, 10], or simply having little or no movement and forming a quasi-static interval, e.g. [11]. In contrast the new Cambrian results from South America, although scattered, suggest a path swinging to the north-east. Results from Africa, particularly Helsley's Cambrian pole at 53°N 34°E, are reasonably consistent with this picture. If, however, Australia and India are positioned in a reconstruction as suggested by Smith and Hallam [8], the single result from the Cambrian of India and the one Ordovician and two Cambrian poles for Australia do not lie on the proposed Gondwanaland path (fig. 7). This is only true if the Smith and Hallam fit is valid; and if the modifications I suggested to their Gondwanaland reconstruction (on the basis of Permo-Triassic results along) [12] are implemented, then the Lower Palaeozoic Indian and Australian results are moved closer to the South American and African results and form a common polar wander path for all of Gondwanaland.

One of the main problems involved in deciphering the Palaeozoic palaeomagnetic results of the northern hemisphere is that of "remagnetization". Creer [1, 13, 14] maintains, despite stability evidence to the contrary [15], that most Lower Palaeozoic and Devonian rocks from Europe and North America have been remagnetized at a later time. However, certain European rocks have been demonstrated by demagnetization studies to have been remagnetized, e.g. [16]. The basic reason for the continued prevalence of the remagnetization hypothesis in the literature is probably simply that magnetic directions obtained from Lower Palaeozoic formations in Europe and North America are similar to directions measured in Carboniferous and Permian rocks. An alternative explanation of the Lower Palaeozoic European palaeomagnetic results is that favoured by Briden [17] which involves an excursion of the pole from a quasi-static position in the mid-Ordovician to mid-Silurian, and return to its former position in the mid-Silurian to early Devonian, combined with some remagnetization of European Lower Palaeozoic results. I prefer a similar explanation to that of Briden, in that the European polar wander path has followed a course in the direction of his excursions, but that it turned around a sharp hairpin bend in the Silurian Period, and that the rate of movement of the pole did not vary as much as envisaged by Briden. So after the Silurian Period the south palaeomagnetic pole returned eastwards passing close to its Cambrian and Ordovician positions. This preferred explanation, of course, implies that remagnetization has not been as extensive as envisaged by Creer.

North American Lower Palaeozoic palaeomagnetic results are scarce and a sensible polar wander path is difficult to construct. Creer again champions an explanation of the results in terms of remagnetization. This interpretation has been debated in the literature, e.g. Roy et al. [18]; comments by Creer [19]; and a reply by Roy et al. [20]. McElhinny and Briden [15] concluded that the North American results were broadly consistent with their Gondwanan polar wander curve, coupled with a position of North America relative to Gondwanaland such as in the computer fit of Bullard et al. [7]. I again prefer an alternative explanation that the North America path is short and diverges to the east of the other paths (fig. 7). This path will undoubtedly require modification when further palaeomagnetic results from North America have been obtained. The tight grouping of Lower Palaeozoic

palaeomagnetic results from Asia, to the west of the three paths described above is shown in fig. 7 and clearly represents the polar wander curve of a separate continental block.

The proposed Lower Palaeozoic polar wander paths do not explain all known palaeomagnetic results; it is simply suggested they are the best approximation, to the real paths, which can be deduced at the present time. The curves are in good agreement with the geological ideas outlined above on the age and origin of the Hercynian and Uralian mountain chains. The polar wander paths of the blocks associated with these orogenies can be seen to converge by the Carboniferous. The separation of land masses across the ocean now represented by the Hercynian and Appalachian mountain belts was about 60° of arc (greater than the size of the present Atlantic) in the Cambrian Period. The palaeomagnetic results, however, do not fit in with the geological knowledge of the history of the Caledonian mountain chain [21]. The most noticeable problems are that (i) there is no hairpin bend in the North American record, corresponding to that in the European and (ii) the Ordovician and Cambrian pole positions for the two blocks lie very close to each other. Neither of these suggests the "proto Atlantic" closed during the Silurian to form the Caledonian mountain chain. Further Lower Palaeozoic results from North America are needed before these apparent contradictions can be explained.

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